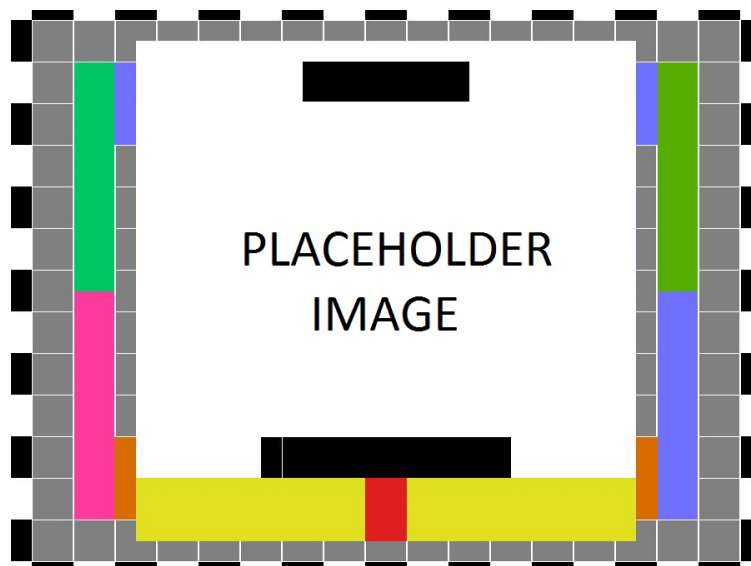


**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE UNIVERSITY OF TEXAS AT ARLINGTON**

**TEST PLAN
CSE 4317: SENIOR DESIGN II
SPRING 2026**



**TEAM NAME
PRODUCT NAME**

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REVISION HISTORY

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1 INTRODUCTION

This Test Plan defines the testing to be performed for Product Name. It is organized as a Test Plan information item per ISO/IEC/IEEE 29119-3:2021 [1].

1.1 PURPOSE

State the purpose of this document and its intended readers (team, customer, instructor).

1.2 SCOPE

State what testing this plan covers (which items, which test levels) and what it does not.

1.3 REFERENCES

Reference the SRS (source of the requirements under test), the ADS/DDS (source of the design under test), and the applicable parts of ISO/IEC/IEEE 29119 [1–4].

2 CONTEXT OF TESTING

2.1 TEST ITEMS

Identify the items to be tested (software builds, firmware, hardware assemblies, integrated system) and their versions/configurations.

2.2 FEATURES TO BE TESTED

List the features/requirements in scope for testing. Reference SRS requirement identifiers.

2.3 FEATURES NOT TO BE TESTED

List features explicitly out of scope for this plan, and why (e.g. deferred/Future requirements).

2.4 ASSUMPTIONS & CONSTRAINTS

State the assumptions and constraints affecting testing (environment availability, hardware lead times, access to the customer site, etc.).

2.5 TEST STAKEHOLDERS

Identify who has an interest in the test results (customer, sponsor, instructor, team) and who approves test completion.

3 TEST RISKS

Record the product risks (risks that the product will not meet requirements) and project risks (risks to the testing effort itself) that drive the testing priorities, per ISO/IEC/IEEE 29119-2 [3].

Table 2: Test risk register (excerpt)

ID	Risk description	Type	Likelihood	Impact
R-01	example product risk	Product	Med	High
R-02	example project risk	Project	Low	Med

4 TEST STRATEGY

4.1 TEST SUB-PROCESSES (LEVELS)

Define the test levels the team will run and the objective of each: unit, integration, system, and acceptance testing. Map each to who performs it and when.

4.2 TEST DESIGN TECHNIQUES

State the techniques used to derive test cases, drawing on ISO/IEC/IEEE 29119-4:2021 [4] — for example equivalence partitioning, boundary value analysis, decision-table testing, state-transition testing, and, where justified, structural (coverage-based) testing.

4.3 TEST COMPLETION (ENTRY/EXIT) CRITERIA

State the entry criteria that must hold before a test level begins and the exit/completion criteria that define “done” (e.g. all Critical/High requirements pass; no open Critical defects; target coverage achieved).

4.4 SUSPENSION & RESUMPTION CRITERIA

State the conditions under which testing is suspended (e.g. a blocking defect) and what must hold to resume.

4.5 REGRESSION & RETESTING

Describe how fixed defects are retested and how regression testing is scoped after changes.

4.6 METRICS

State the metrics collected (requirements coverage, test pass rate, open-defect counts by severity) and the reporting cadence.

5 TEST DELIVERABLES

List the documents and artifacts produced: this Test Plan, test case specifications, test data, test scripts, test execution logs/results, defect reports, and the test completion report.

6 TEST ENVIRONMENT & TOOLING

Describe the hardware, software, network, and data needed to execute the tests, and the tools used (test runners, CI, hardware-in-the-loop rigs, measurement instruments).

7 ROLES, RESPONSIBILITIES & STAFFING

Identify who is responsible for test design, execution, and sign-off, and any training required.

8 SCHEDULE & ESTIMATES

Provide the testing schedule aligned to the sprint plan and the effort estimates for the testing tasks.

9 TEST CASE SPECIFICATIONS

Specify each test case using the fields below, adapted from the test case specification of ISO/IEC/IEEE 29119-3 [1]. Repeat per test case.

9.1 TC-01: [TEST CASE TITLE]

- **Objective:** what this test verifies
- **Traces to:** SRS requirement ID(s) and design element(s)
- **Preconditions:** required state/setup before execution

- **Inputs:** test data / actions
- **Expected results:** the measurable, observable pass condition
- **Test technique:** the 29119-4 technique used to derive the case
- **Priority:** Critical / High / Moderate / Low

10 REQUIREMENTS TRACEABILITY

Provide a matrix linking every requirement under test to the test case(s) that verify it, closing the traceability chain begun in the SRS and continued through the ADS and DDS.

Table 3: Requirement-to-test-case traceability (excerpt)

Req ID	Requirement (abbreviated)	Test case(s)
FR-01	abbreviated requirement text	TC-01, TC-02
PE-01	abbreviated requirement text	TC-05

REFERENCES

- [1] ISO/IEC/IEEE, “ISO/IEC/IEEE 29119-3:2021 — Software and systems engineering — Software testing — Part 3: Test documentation,” International Organization for Standardization, Geneva, Switzerland, Tech. Rep., 2021.
- [2] —, “ISO/IEC/IEEE 29119-1:2022 — Software and systems engineering — Software testing — Part 1: General concepts,” International Organization for Standardization, Geneva, Switzerland, Tech. Rep., 2022.
- [3] —, “ISO/IEC/IEEE 29119-2:2021 — Software and systems engineering — Software testing — Part 2: Test processes,” International Organization for Standardization, Geneva, Switzerland, Tech. Rep., 2021.
- [4] —, “ISO/IEC/IEEE 29119-4:2021 — Software and systems engineering — Software testing — Part 4: Test techniques,” International Organization for Standardization, Geneva, Switzerland, Tech. Rep., 2021.